

## Contents

- 1 字串
  - 1.1 字典樹 trie . . . . .
  - 1.2 KMP . . . . .
- 2 最短路徑
  - 2.1 basic . . . . .
- 3 圖論
  - 3.1 凸包問題 . . . . .
- 4 數學
  - 4.1 公式 . . . . .

## 1 字串

### 1.1 字典樹 trie

```

1 struct trie{
2     trie *nxt[26]={};
3     int cnt=0;
4     int sz=0;
5 };
6 trie *root=new trie();
7 void insert(const string &s){
8     trie *now=root;
9     for(auto i:s){
10        if(now->nxt[i-'a']==NULL){
11            now->nxt[i-'a']=new trie();
12        }
13        now=now->nxt[i-'a'];
14    }
15    int qry(string &s){
16        trie *now=root;
17        for(auto i:s){
18            if(now->nxt[i-'a']==NULL) return;
19            now=now->nxt[i-'a'];
20        }
21        return now->cnt;
22    }
23 }
```

### 1.2 KMP

```

1 vector<int> kmp(string s,string t){
2     if(t.size()>s.size()) return {};
3     vector<int> pi(t.size(),0);
4     for(int i=1,j=0;i<t.size();i++){
5         while(j>0 && t[j]!=t[i]){
6             j=pi[j-1];
7         }
8         if(t[j]==t[i]) j++;
9         pi[i]=j;
10    }
11    vector<int> ans;
12    for(int i=0,j=0;i<s.size();i++){
13        while(j>0 && t[j]!=s[i]){
14            j=pi[j-1];
15        }
16        if(t[j]==s[i]) j++;
17
18        if(j==t.size()){
19            ans.push_back(i-j+1);
20            j=pi[j-1];
21        }
22    }
23    return ans;
24 }
```

## 2 最短路徑

### 2.1 basic

```

1 // c++ code
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 int main() {
6     // test comment
7     cout << "test string\n";
8 }
```

## 3 圖論

### 3.1 凸包問題

```

1 struct node{
2     int x,y;
3     bool operator<(const node &other) const{
4         return
5             (x<other.x) || (x==other.x && y<other.y);
6     };
7 };
8 struct ConvexHull{
9     vector<node> vec;
10    int n;
11    void init(vector<node> &v){
12        vec=v;
13        n=v.size();
14    }
15    int cross(node a,node o,node b){
16        return
17            (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
18    }
19    vector<int> point;
20    vector<int> area;
21    void build(){
22        point=vector<int>(n+1);
23        area=vector<int>(n+1);
24        vector<node> up,down;
25        int upsum=0;
26        int downsum=0;
27        for(int i=0;i<n;i++){
28            while(up.size()>=2 &&
29                cross(up[up.size()-2],
30                    up[up.size()-1],
31                    vec[i])<=0){
32                upsum-=cross(up[up.size()-2],
33                            {0,0},up[up.size()-1]);
34                up.pop_back();
35            }
36            up.push_back(vec[i]);
37            if(up.size()>1)
38                upsum+=cross(up[up.size()-2],
39                            {0,0},
40                            up[up.size()-1]);
41
42            while(down.size()>=2 &&
43                cross(down[down.size()-2],
44                    down[down.size()-1],
45                    vec[i])>=0){
46                downsum-=cross(down[down.size()-2],
47                            {0,0},down[down.size()-1]);
48                down.pop_back();
49            }
50            down.push_back(vec[i]);
51            if(down.size()>1)
52                downsum+=cross(down[down.size()-2],
53                            {0,0},down[down.size()-1]);
54
55            area[i+1]=abs(upsum-downsum);
56
57            if(i+1==1) point[i+1]=1;
58            else point[i+1]=up.size()+down.size()-2;
59        }
60    }
61    double getarea(int p){
62        return area[p]/2.0;
63    }
64    int getnum(int p){
65        return point[p];
66    }
67 }
```

## 4 數學

### 4.1 公式

- 中文測試

$$\cdot \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$